Conservation, Energy Geometry, and Reversibility

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Definition: First integral

Let $U \subseteq \mathbb{R}^2$ be open and let $\mathbf{F} : U \to \mathbb{R}^2$ be continuously differentiable. A continuously differentiable function $E : U \to \mathbb{R}$ is a **first integral** of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ if

$$\nabla E(\mathbf{z}) \cdot \mathbf{F}(\mathbf{z}) = 0$$

for every $\mathbf{z} \in U$. Equivalently, $E(\mathbf{z}(t))$ is constant along every solution.

Definition: Conservative system

A system on an open set U is **conservative** if it has a first integral that is not constant on any nonempty open subset of U.

Question: Why is mechanical energy conserved?

Let $I \subseteq \mathbb{R}$ be an open interval, let $V \in C^2(I)$, let $m > 0$, and suppose $x(t) \in I$ satisfies

$$m\ddot{x} = F(x), \qquad F(x) = -V'(x).$$

Find a first integral of the planar system obtained with $y = \dot{x}$.

Solution. The planar system is

$$\dot{x} = y, \qquad \dot{y} = -\frac{1}{m}V'(x).$$

Multiplying the second-order equation by $\dot{x}$ gives

$$m\dot{x}\ddot{x} + V'(x)\dot{x} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}m\dot{x}^2 + V(x)\right) = 0.$$

Thus

$$E(x, y) = \frac{1}{2}my^2 + V(x)$$

is conserved. Directly,

$$\mathbf{G}(x, y) = \left(y, -\frac{1}{m}V'(x)\right), \qquad \nabla E \cdot \mathbf{G} = (V'(x), my) \cdot \left(y, -\frac{1}{m}V'(x)\right) = 0.$$

Proposition: A conservative system has no attracting equilibrium

A conservative system has no attracting equilibrium.

Proof. Let E be a first integral that is not constant on any nonempty open set. Suppose $\mathbf{z}^*$ is attracting. Its basin contains a nonempty open neighborhood of $\mathbf{z}^*$. For any $\mathbf{z}_0$ in the basin, Conservation and continuity give

$$E(\mathbf{z}_0) = E(\Phi_t(\mathbf{z}_0)) \longrightarrow E(\mathbf{z}^*) \qquad (t \to \infty).$$

Hence $E(\mathbf{z}_0) = E(\mathbf{z}^*)$ for every $\mathbf{z}_0$ in the basin. Thus E is constant on a nonempty open set, contradicting the definition of a conservative system.

Visualization: Potential and phase-plane energy level

Let $I \subseteq \mathbb{R}$ be an interval, let $m > 0$, let $V \in C^2(I)$, and consider

$$\dot{x} = y, \qquad \dot{y} = -\frac{1}{m}V'(x), \qquad E(x,y) = \frac{1}{2}my^2 + V(x).$$

For

$$E(x,y) = \frac{1}{2}my^2 + V(x) = h,$$

the two velocity branches are

$$y = \pm\sqrt{\frac{2}{m}\big(h - V(x)\big)}.$$

Therefore

potential condition	phase-plane meaning
$V(x) < h$	two states with velocities $\pm\lvert y\rvert$
$V(x) = h,\ V'(x) \neq 0$	$y = 0$; non-equilibrium turning point
$V(x) = h,\ V'(x) = 0$	equilibrium
$V(x) > h$	no state on the energy level
$V'(x^*) = 0,\ V''(x^*) > 0$	nearby oval levels about $(x^*, 0)$
$V'(x^*) = 0,\ V''(x^*) < 0$	saddle energy at $(x^*, 0)$

Since $y = \dot{x}$, the upper branch moves right and the lower branch moves left.

Definition: Regular level

Let $U \subseteq \mathbb{R}^2$ be open, let $E \in C^1(U)$, and let $h \in \mathbb{R}$. The level set

$$\{p \in U : E(p) = h\}$$

is **regular** if

$$\nabla E(p) \neq \mathbf{0}$$

at every point p in that level set.

Lemma: Definite Hessian gives closed nearby levels

Let E be twice continuously differentiable near $\mathbf{z}^*$, suppose

$$\nabla E(\mathbf{z}^*) = \mathbf{0},$$

and suppose $D^2E(\mathbf{z}^*)$ is positive definite. Then $\mathbf{z}^*$ is a strict local minimum, and for all sufficiently small $c > 0$ the nearby component of

$$E(\mathbf{z}) = E(\mathbf{z}^*) + c$$

is a regular simple closed curve surrounding $\mathbf{z}^*$. For a negative definite Hessian, the same conclusion holds with $c < 0$.

Proof. Assume the Hessian is positive definite. By continuity, there are a radius $R > 0$ and constants $0 < m \le M$ such that

$$m\|\mathbf{v}\|^2 \le \mathbf{v}^{\mathsf{T}} D^2E(\mathbf{z})\mathbf{v} \le M\|\mathbf{v}\|^2$$

whenever $\|\mathbf{z} - \mathbf{z}^*\| \le R$. For a unit vector $\mathbf{e} \in S^1$, define

$$q_{\mathbf{e}}(r) = E(\mathbf{z}^* + r\mathbf{e}) - E(\mathbf{z}^*).$$

Because $\nabla E(\mathbf{z}^*) = \mathbf{0}$,

$$q'_{\mathbf{e}}(r) = \int_0^r \mathbf{e}^{\mathsf{T}} D^2E(\mathbf{z}^* + s\mathbf{e})\mathbf{e}\, ds,$$

so, for $0 < r \le R$,

$$mr \le q'_{\mathbf{e}}(r) \le Mr, \qquad \frac{m}{2}r^2 \le q_{\mathbf{e}}(r) \le \frac{M}{2}r^2.$$

Thus, for every sufficiently small $c > 0$ and every $\mathbf{e} \in S^1$, there is a unique radius $r_c(\mathbf{e}) \in (0, R)$ satisfying

$$E(\mathbf{z}^* + r_c(\mathbf{e})\mathbf{e}) = E(\mathbf{z}^*) + c.$$

The implicit-function theorem makes r_c continuously differentiable in $\mathbf{e}$. Hence

$$\mathbf{e} \longmapsto \mathbf{z}^* + r_c(\mathbf{e})\mathbf{e}$$

parametrizes a simple closed curve surrounding $\mathbf{z}^*$. Moreover, $q'_{\mathbf{e}}(r_c(\mathbf{e})) > 0$, so ∇E does not vanish on this curve; the level is regular. Replacing E by $-E$ proves the negative-definite case. $\square$

Proposition: Regular closed energy levels are periodic orbits

Let E be a continuously differentiable first integral of a continuously differentiable planar vector field $\mathbf{F}$. Let C be a regular simple closed component of a level set of E. If C contains no equilibrium, then C is a periodic orbit.

Proof. Conservation makes $\mathbf{F}$ tangent to C. Since C is compact and contains no equilibrium, $\|\mathbf{F}\|$ has a positive minimum on C. Thus a trajectory on C traverses the curve and returns to its initial point in finite time. Uniqueness makes the trajectory periodic.

Definition: Nonlinear center

An isolated equilibrium $\mathbf{z}^*$ of a planar system is a **nonlinear center** if every sufficiently nearby non-equilibrium point lies on a periodic orbit that is a simple closed curve surrounding $\mathbf{z}^*$.

Theorem: Definite-Hessian nonlinear-center criterion

Let $\mathbf{F}$ be continuously differentiable near an isolated equilibrium $\mathbf{z}^*$ of a planar system, and let E be a twice continuously differentiable first integral. Suppose

$$\nabla E(\mathbf{z}^*) = \mathbf{0}$$

and $D^2E(\mathbf{z}^*)$ is positive definite or negative definite. Then every non-equilibrium point sufficiently near $\mathbf{z}^*$ lies on a periodic orbit that is a simple closed level curve of E surrounding $\mathbf{z}^*$. Hence $\mathbf{z}^*$ is a nonlinear center.

Proof. Assume the Hessian is positive definite; otherwise replace E by $-E$. The radial construction in the preceding lemma shows that a sufficiently small punctured neighborhood is foliated by regular simple closed level curves surrounding $\mathbf{z}^*$. Since $\mathbf{z}^*$ is isolated, the neighborhood can be chosen to contain no other equilibrium. The preceding proposition makes each of these level curves a periodic orbit.

Question: Why is a non-strict local minimum insufficient?

Consider

$$\dot{x} = 0, \qquad \dot{y} = x^2 + y^2, \qquad E(x, y) = x^2.$$

Verify that the origin is an isolated equilibrium and a local minimum of a first integral that is not constant on any nonempty open set, but is not a center.

Solution. The equilibrium equations are

$$0 = 0, \qquad x^2 + y^2 = 0,$$

so the origin is the unique equilibrium. Also,

$$\dot{E} = 2x\dot{x} = 0,$$

and $E = x^2 \geq 0 = E(0, 0)$. The minimum is not strict, and the nearby levels $x^2 = c$ are pairs of vertical lines rather than closed curves. At every non-equilibrium point,

$$\dot{y} = x^2 + y^2 > 0,$$

so no non-equilibrium trajectory is periodic. The missing hypothesis is the closed-contour condition; a definite Hessian would guarantee it.

Finally, E is not constant on any nonempty open set: every open disk contains a horizontal line segment on which x^2 is not constant.

Definition: Homoclinic trajectory

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \to \mathbb{R}^2$, and let $p \in U$ be an equilibrium of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$. A nonconstant solution $\mathbf{z} : \mathbb{R} \to U$ is **homoclinic to** p if

$$\lim_{t\to-\infty} \mathbf{z}(t) = p = \lim_{t\to\infty} \mathbf{z}(t).$$

Its image is a **homoclinic trajectory** or **homoclinic orbit**.

Question: What is the complete phase portrait of a double well?

Consider

$$\ddot{x} = x - x^3.$$

Find the equilibria, conserved energy, local classifications, and energy-level regimes.

Solution. With $y = \dot{x}$,

$$\dot{x} = y, \qquad \dot{y} = x - x^3.$$

Choose

$$V(x) = -\frac{1}{2}x^2 + \frac{1}{4}x^4, \qquad E(x,y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

The equilibria are

$$(0,0), \qquad (1,0), \qquad (-1,0).$$

The Jacobian is

$$J(x,y) = \begin{pmatrix} 0 & 1 \\ 1-3x^2 & 0 \end{pmatrix}.$$

At $(0,0)$, the eigenvalues are ± 1, so it is a saddle. At $(\pm 1, 0)$, the eigenvalues are $\pm i\sqrt{2}$, and

$$D^2E(\pm 1, 0) = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

is positive definite. Hence $(\pm 1, 0)$ are nonlinear centers.
Their energy is $-1/4$ and the saddle energy is 0. On $E = h$,

$$y = \pm\sqrt{2h + x^2 - \frac{1}{2}x^4}.$$

Therefore

h	level-set geometry
$h < -1/4$	empty
$h = -1/4$	the two center equilibria
$-1/4 < h < 0$	two disjoint periodic ovals
$h = 0$	the saddle together with two homoclinic-orbit closures
$h > 0$	one periodic oval surrounding all three equilibria

The two nonconstant homoclinic trajectories on $h = 0$ are not periodic because approach to the saddle takes infinite time.

Visualization: Energy surface for the double well

For

$$w = E(x,y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

the critical points and heights are

(x,y)	$E(x,y)$	surface type
$(-1,0)$	$-1/4$	minimum
$(1,0)$	$-1/4$	minimum
$(0,0)$	0	saddle

Horizontal slices satisfy

$h < -1/4$	empty
$h = -1/4$	the two minima
$-1/4 < h < 0$	two ovals
$h = 0$	figure-eight saddle-energy level
$h > 0$	one outer oval

and project to the corresponding phase-plane energy levels.

Definition: Heteroclinic trajectory

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \to \mathbb{R}^2$, and let $p, q \in U$ be distinct equilibria of $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$. A solution $\mathbf{z} : \mathbb{R} \to U$ is **heteroclinic from p to q** if

$$\lim_{t\to-\infty} \mathbf{z}(t) = p, \qquad \lim_{t\to\infty} \mathbf{z}(t) = q.$$

Its image is a **heteroclinic trajectory** or **heteroclinic orbit**.

Observation: Saddle connections are not periodic

Every homoclinic or heteroclinic solution has an equilibrium limit as $t \to \infty$. A periodic solution that has a limit as $t \to \infty$ must be constant: periodicity repeats every point on its trajectory arbitrarily far in the future, so every such point must equal the limit. A homoclinic solution is nonconstant by definition, and a heteroclinic solution has two distinct limiting equilibria. Therefore neither trajectory is periodic.

Recall: Involution

Let U be a set. A map $R : U \to U$ is an **involution** if

$$R(R(\mathbf{z})) = \mathbf{z}.$$

Recall: Reversing symmetry

Let $U \subseteq \mathbb{R}^2$ be open, let $\mathbf{F} : U \to \mathbb{R}^2$, and let $R : U \to U$ be a differentiable involution. The map R is a **reversing symmetry** for $\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z})$ if

$$\mathbf{F}(R(\mathbf{z})) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z}).$$

Observation: A reverser reverses solution arcs

If R is a reversing symmetry and $\mathbf{z}(t)$ is a solution, then $R(\mathbf{z}(-t))$ is also a solution.

Observation: Reflection-reversibility test

For reflection across the x-axis,

$$R(x, y) = (x, -y).$$

Writing $\mathbf{F} = (f, g)$, reversibility becomes

$$f(x, -y) = -f(x, y), \qquad g(x, -y) = g(x, y).$$

Proposition: Undamped one-dimensional mechanics is reversible

Let $I \subseteq \mathbb{R}$ be open, let $m > 0$, and let $F : I \to \mathbb{R}$ be continuous. The system on $I \times \mathbb{R}$

$$\dot{x} = y, \qquad \dot{y} = \frac{1}{m}F(x)$$

is reversible under $R(x, y) = (x, -y)$.

Proof. Here

$$\mathbf{F}_{\text{sys}}(x, y) = \left(y, \frac{1}{m}F(x)\right), \qquad DR = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Therefore

$$\mathbf{F}_{\text{sys}}(R(x, y)) = \left(-y, \frac{1}{m}F(x)\right) = -DR\mathbf{F}_{\text{sys}}(x, y).$$

Theorem: Nonlinear centers in reflection-reversible systems

Let $U \subseteq \mathbb{R}^2$ be a reflection-invariant open neighborhood of the origin, and let $\mathbf{F} : U \to \mathbb{R}^2$ be continuously differentiable. Suppose the origin is an equilibrium, the system is reversible under $R(x, y) = (x, -y)$, and $D\mathbf{F}(0, 0)$ has eigenvalues $\pm i\omega$ with $\omega > 0$. Then every sufficiently small non-equilibrium trajectory is closed; the origin is a nonlinear center.

Proof sketch. Reversibility forces the linearization to have the form

$$J = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}, \qquad ab = -\omega^2 < 0.$$

After reversing time if necessary, take $a > 0 > b$. On a punctured neighborhood of the origin, define polar variables by

$$r^2 = x^2 + y^2, \qquad \theta = \arg(x + iy).$$

Theorem: Nonlinear centers in reflection-reversible systems (continued)

Then

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = \frac{bx^2 - ay^2 + o(r^2)}{r^2} \qquad (r \to 0).$$

Let $\mu = \min\{-b, a\} > 0$. Then

$$bx^2 - ay^2 \le -\mu r^2.$$

Because $o(r^2)/r^2 \to 0$ as $r \to 0$, throughout a sufficiently small punctured neighborhood,

$$\dot{\theta} \le -\frac{\mu}{2} < 0.$$

Choose a disk of radius R on which the angular bound holds and

$$\|\mathbf{F}(\mathbf{z})\| \le C\|\mathbf{z}\|.$$

The bound $|\dot{\theta}| \ge \mu/2$ shows that an angular change of 2π takes at most $4\pi/\mu$. Grönwall's inequality gives

$$\|\mathbf{z}(t)\| \le \|\mathbf{z}(0)\|e^{C|t|}.$$

Thus an initial radius smaller than $Re^{-4\pi C/\mu}$ keeps the trajectory inside the disk long enough to reach two consecutive crossings of $y = 0$. Their angular separation is π. Reflecting the arc between the crossings and reversing its arrows produces the continuation from the second crossing back to the first. Uniqueness closes the orbit and prevents self-intersections. The loop surrounds the origin because its angle changes by 2π. Finally, $\det J = \omega^2 > 0$, so the inverse-function theorem makes the origin an isolated equilibrium.

Question: Does reversibility turn a linear center into a nonlinear center?

Consider

$$\dot{x} = y - y^3, \qquad \dot{y} = -x - y^2.$$

Find the equilibria and classify them.

Solution. The first component is odd in y, and the second is even in y, so $R(x, y) = (x, -y)$ is a reverser. At the origin,

$$D\mathbf{F}(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

whose eigenvalues are $\pm i$. The reversible-center theorem makes the origin a nonlinear center.
The equilibrium equations are

$$y(1 - y^2) = 0, \qquad -x - y^2 = 0.$$

Thus the other equilibria are $(-1, 1)$ and $(-1, -1)$. Since

$$D\mathbf{F}(x, y) = \begin{pmatrix} 0 & 1 - 3y^2 \\ -1 & -2y \end{pmatrix},$$

the determinant at either point is $-2 < 0$. Both are saddles.

Question: Where is the homoclinic loop?

Consider
$$\dot{x} = y, \qquad \dot{y} = x - x^2.$$
Find an exact energy equation for the homoclinic loop in $x \geq 0$.

Solution. The origin is a saddle, and
$$E(x,y) = \frac{1}{2}y^2 - \frac{1}{2}x^2 + \frac{1}{3}x^3$$
is conserved because
$$\dot{E} = y(x - x^2) + (-x + x^2)y = 0.$$
The saddle energy is $E = 0$. Hence the upper branch satisfies
$$y = x\sqrt{1 - \frac{2x}{3}}, \qquad 0 < x \leq \frac{3}{2},$$
and reaches the x-axis at $(3/2, 0)$. Reflection across the x-axis with time reversal gives the lower branch
$$y = -x\sqrt{1 - \frac{2x}{3}}, \qquad 0 < x \leq \frac{3}{2}.$$
The two branches approach the origin as $t \to \pm\infty$ and form a homoclinic loop. More precisely, the upper branch leaves the origin as $t \to -\infty$ and reaches $(3/2, 0)$ in finite time; the solution then follows the lower branch and approaches the origin as $t \to +\infty$.

Question: Does reversibility imply conservation?

Consider
$$\dot{x} = -2\cos x - \cos y, \qquad \dot{y} = -2\cos y - \cos x$$
on $\mathbb{R}^2$. The vector field is 2π-periodic in each coordinate. Show that the system is reversible but not conservative.

Solution. With $R(x,y) = (-x,-y)$, $DR = -I$, and cosine even,
$$\mathbf{F}(R\mathbf{z}) = \mathbf{F}(\mathbf{z}) = -DR\mathbf{F}(\mathbf{z}),$$
so the system is reversible. In the representative cell $[-\pi, \pi) \times [-\pi, \pi)$, its equilibria are
$$(\sigma\pi/2, \rho\pi/2), \qquad \sigma, \rho \in \{-1, 1\}.$$
The Jacobian is
$$D\mathbf{F}(x,y) = \begin{pmatrix} 2\sin x & \sin y \\ \sin x & 2\sin y \end{pmatrix}.$$
At $(-\pi/2, -\pi/2)$, the eigenvalues are $-3, -1$, so it is an asymptotically stable node and therefore has an open attracting basin. A conservative system cannot have an open attracting basin. Therefore this reversible system is not conservative.

Visualization: Conservation and reversibility

structure	identity	consequence
first integral	$\nabla E \cdot \mathbf{F} = 0$	trajectories remain on level sets
reverser	$\mathbf{F}(R\mathbf{z}) = -DR(\mathbf{z})\mathbf{F}(\mathbf{z})$	reflected arcs carry reversed time
both	two independent checks	energy geometry plus orbit symmetry

Observation: Purely imaginary eigenvalues are inconclusive

Purely imaginary eigenvalues of the Jacobian do not, by themselves, imply the existence of a first integral, a reversing symmetry, or a nonlinear center.